

Characterization of Adjacent-Track Interference Using Repeated-Write THMap for High Track Density in HAMR

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In heat-assisted magnetic recording (HAMR), adjacent-track interference (ATI) remains a critical challenge that limits the achievable track density. In this study, ATI is evaluated at the drive-level by deriving the optimum track pitch following repeated adjacent-track write operations. The optimum track pitch increases logarithmically with an increasing number of repeated adjacent-track writes, indicating the cumulative nature of ATI. Magnetization switching probability under repeated write conditions is characterized using THMap, which reveals a logarithmic decrease in the average write temperature, T_w , with increasing write cycles. Track-edge propagation estimated from THMap measurements is consistent with the change in optimum track pitch, suggesting that ATI is primarily governed by track-edge behavior. These results highlight the critical role of the cross-track thermal gradient and the importance of minimizing ΔT_w , defined as the reduction in T_w induced by repeated writes, to improve ATI tolerance in high track density HAMR systems.

Index Terms—heat-assisted magnetic recording (HAMR), adjacent-track interference (ATI), THMap.

I. INTRODUCTION

HEAT-ASSISTED MAGNETIC RECORDING (HAMR) is a key technology for enabling areal density capabilities (ADC) beyond 4 Tbit/in² in hard disk drives (HDDs) [1]. While HAMR allows the aggressive scaling of track density by reducing the magnetic coercivity during write operation through localized heating, adjacent-track interference (ATI), which manifests as on-track degradation induced by repeated write operations on neighboring tracks, remains a major obstacle to achieving ultra-high track density. Therefore, recording formats are designed to ensure sufficient tolerance against ATI. For example, in an industry-standard approach for conventional magnetic recording (CMR) systems, format selection criteria assume tolerance to six repeated adjacent-track writes [2].

In HAMR, the impact of ATI has primarily been discussed in terms of written-in signal quality such as signal-to-noise ratio (SNR) [3], [4]. While ATI is also expected to affect track pitch selection and consequently ADC, systematic drive-level evaluations of ATI from the perspective of track pitch and ADC optimization remain limited.

Reducing ATI requires understanding how ATI develops and identifying head- and media-related factors that contribute to its behavior. Previous work has suggested that ATI mainly depends on the cumulative degradation of written-in patterns following repeated adjacent-track writes and the ability of grains to retain their magnetization under such conditions [5]. Simulation results from this study demonstrated that this framework can explain ATI trends across media with different thermal stabilities, highlighting the role of media-related factors in ATI behavior. Experimentally, ATI has also been shown to exhibit cumulative behavior with increasing writes [3], [4]. However, while the cumulative nature of ATI has been recognized, a systematic drive-level investigation demonstrating how magnetization switching evolves under repeated writes and how

this evolution quantitatively links to ATI behavior has not yet been reported.

In this study, a drive-level evaluation is carried out using THMap to quantify the media temperature dependence of magnetization switching probability and its evolution under repeated writes [6]. ATI is evaluated in terms of optimum track pitch for HDDs with various combinations of heads and media designs. The evolution of ATI under repeated adjacent-track writes is examined alongside the corresponding changes in magnetization switching probability. The results clarify the cumulative nature of magnetization switching probability and establish its quantitative relationship to ATI, providing insight relevant to the design of ATI-tolerant heads and media, and system-level applications for HAMR HDDs.

II. EXPERIMENTS

All experiments were performed using functional development-stage HAMR HDDs with multiple heads and media designs. The drives were operated at a rotational speed of 7200 rpm and an ambient temperature of approximately 35 °C. The laser power and write coil current were individually optimized for each head. The cross-track thermal gradient was estimated from the media temperature dependence of the write width, where the media temperature was converted from the laser current using THMap measurements [6].

At a fixed linear density of 2167 kFCI (kilo flux changes per inch), ATI performance was evaluated by determining the optimum track pitch that gives 50% sector failure rate after an arbitrary number of adjacent-track writes at each side. The track pitch increases with increasing adjacent-track write cycles. Taking the track pitch after a single adjacent-track write as a reference, the increase in track pitch with repeated write cycles (6, 20, 50 cycles) was utilized as an indicator of ATI performance, defined as Δp . Cross-track read position was optimized for each write condition.

The average write temperature (T_w), defined as the media temperature at which 50 % of the grains switch their magnetization to the head-field direction, was measured using the THMap [6]. THMap estimates magnetization switching probability as a function of media temperature by measuring noise power after DC-polarity band writes. In this study, we repeated the DC-polarity band writes up to 50 cycles to determine T_w change (ΔT_w) as a function of write cycles.

III. RESULTS AND DISCUSSION

In ATI evaluations based on track pitch measurements after repeated adjacent-track writes, Δp increased logarithmically with increasing write cycles. Similarly, THMap measurements showed that T_w decreased logarithmically with increasing write cycles. This indicates that repeated writes enhance magnetization switching probability under identical media temperature conditions, as quantified by THMap. Such logarithmic trends are consistent with reported logarithmic SNR loss and write width enhancement, indicating cumulative magnetization switching that results in ATI [3], [4].

Fig. 1 shows a schematic illustration of track-edge propagation (Δx) after repeated writes. In this model, Δx is calculated as ΔT_w divided by the cross-track thermal gradient. Δx with increasing write cycles was estimated and compared with Δp to evaluate the contribution of Δx to ATI performance. Fig. 2 shows the relationship between Δx and Δp after 50 write cycles. Δx and Δp showed a correlation with R^2 of 0.49, demonstrating that ATI is closely related to magnetization switching enhancement by repeated writes. Moreover, Δp scaled approximately with Δx , indicating that ATI is dominated by the track-edge propagation caused by the elevated adjacent-track temperature at the track-edge, as suggested in the previous study [4].

These results reinforce the importance of heads and media designs with a large cross-track thermal gradient. In addition, minimizing ΔT_w is also important, as it primarily reflects the magnetization switching characteristics of the media.

IV. SUMMARY

ATI performance was evaluated at the drive-level by measuring the optimum track pitch after repeated adjacent-track writes using HAMR HDDs. Logarithmic behavior was observed in both the increase in the optimized track pitch (Δp) and the decrease in T_w (ΔT_w) measured by THMap.

The relationship between ATI and magnetization switching probability under repeated writes was examined by comparing Δp and Δx , estimated from ΔT_w . The observed correlation between Δx and Δp suggests a quantitative link between the evolution of magnetization switching probability and ATI behavior. In addition, the increase in Δp after adjacent-track writes was found to scale with Δx , indicating that ATI mainly affects the track-edge where the media temperature during adjacent write is high.

These results underscore the importance of heads and media designs with a high thermal gradient and minimal ΔT_w for ATI performance, suggesting that further ADC improvement can be achieved in ATI-limited regimes.

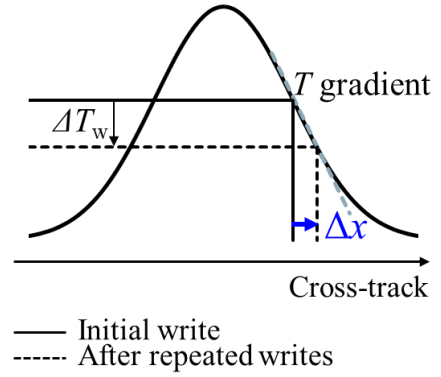


Fig. 1. Schematic illustration of the media temperature profile along the cross-track direction and the resulting track-edge propagation (Δx) after repeated adjacent-track writes. T_w decreases with repeated write cycles. The corresponding Δx is estimated from the decrease in T_w (ΔT_w) and the cross-track thermal gradient.

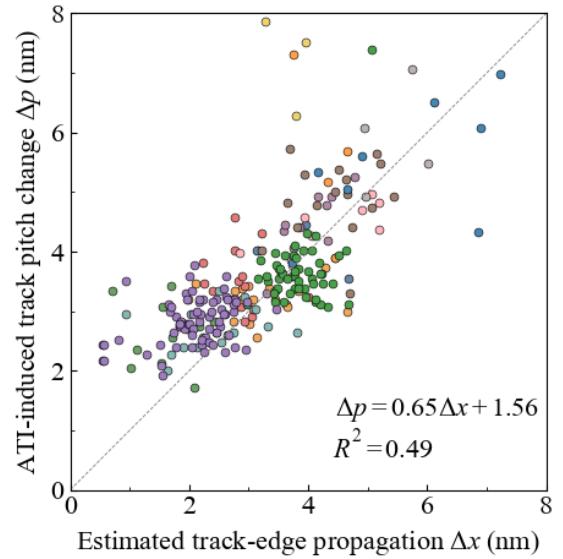


Fig. 2. Relationship between the estimated track-edge propagation (Δx) and ATI-induced track pitch change (Δp) after 50 repeated write cycles, measured at the inner-diameter region of the disk under skewed conditions. Different colors indicate different head-media combinations. The dashed line represents the one-to-one relation ($\Delta p = \Delta x$). A linear regression yields a slope of 0.65 and an intercept of 1.56 nm ($R^2 = 0.49$), suggesting that Δp scales approximately with Δx .

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